

Towards Word Sense Disambiguation System for Maithili Language

1st Jitendra Singh Solanki
Computer Science and Engineering
Mandsaur University
Mandsaur, India
jitendrasinghsolanki306@gmail.com

2nd Nidhi
Computer Science and Engineering
Mandsaur University
Mandsaur, India
nidhisharmacbs@gmail.com

3rd Dr. Prabhat Kumar Singh
Computer Science and Engineering
Mandsaur University
Mandsaur, India
prabhatprobable@gmail.com

Abstract—Word Sense Disambiguation (WSD) is a significant exercise in the field of Natural Language Processing(NLP) which is used to determine the appropriate meaning of words in their context. Maithili is an Indo-Aryan language whose digital resources and tools of language technology are limited as the language is spoken by millions of people. Maithili words are not very few with two or more meanings and without proper resources it is hard to select the right meaning with the help of computers. This review article discusses the issues and strategies to execute Word Sense Disambiguation (WSD) in Maithili. The absence of annotated datasets, digital dictionaries, and NLP tools are the primary issues that we are going to discuss. In the paper, the author discusses various ways that WSD can be solved using rule based, machine learning, and deep learning models among others applied to Maithili. We also introduce a combination of linguistic literacy and current computational solutions to create successful WSD systems in Maithili. It demonstrates that Word Sense Disambiguation of Maithili can be made and it can be valuable to numerous applications of the language, such as machine translation, information retrieval, and sentiment analysis, with the right data collection methods, annotation, and the application of advanced algorithms. This review is meant to inform further research and development in computing linguistics of the Maithili language.

Index Terms—Word Sense Disambiguation(WSD), Maithili, Natural Language Processing NLP, Machine Learning, Deep Learning

I. INTRODUCTION

Natural Language Processing(NLP) today is not just a branch of Artificial Intelligence(AI); It's a revolutionary technology that gives machines the power to understand human emotions,context and conversation ,respond intelligently,and analyze data in completely new ways.Imagine how you're reading these words right now-your mind quickly understands every word,the overall message,even the little hints and feelings hidden behind the sentences .For a computer,though ,this is almost impossible, because at the end of the day, computers just see sequences of ones and zeros. Here's whrer Natural Language Processing steps in: it's the bridge that turns our language into something machines can process and understand .Think about your daily life, Whenever you ask your phone's assistant

a question use grammer check in a document,or translate a text into another language,you're actually using NLP! These features have become so everyday that it's easy to forget the incredibly complex Science working behind the scenes. When a computer reads a sentence, it has to Split the text into smaller pieces(like individual words or sentences), Figure out how those words fit together, work out the meaning of each word,depending on how it's used , create its own answers,summaries or even new translations.Every one of these steps comes with its own set of challenges,and researchers are constantly inventing clever new ways to help computers get better and better at understanding language.Next time you chat with a bot or translate a phrase online ,take a second to appreciate just how much is happening beneath the surface.

Maithili is a fascinating language with a rich history and culture significance.It belongs to the Indo Aryan language family ,which includes many languages spoken across south Asia.The language is primarily spoken in the Mithili region, which covers northern Bihar,parts of Jharkhand in India,and the Terai region of southern Nepal.In the modern world,computers can understand and process some languages, like English, much better than others.when a language does not have enough digital data,tools or computer programs to help reseachers,we call it “under-resourced “ in Natural Language Processing.This does not mean the language is less valuable or less complex-it just means that computers don't have enough materials to work with that language easily.For a big language like English , there are huge collections of books, articals , and websites.Researchers have built big databases and lists, like Wordnet, which help computers understand the meaning and relationships of words, but Maithili is different. Even though Maithili has a long history and a rich culture,most of its literature is still only found in printed books or shared by speaking, not in digital form. There are very few digital dictionaries or lists of Maithili. Because of this, it is much harder to build computer programs that can understand or process Maithili. This is what we mean when we say Maithili is “under-resourced” in NLP. Among all the challenges in NLP, Word sense Disambiguation stands out as particularly important yet

difficult. WSD is the task of determining which meaning of a word is being used in a particular context when that word has multiple possible meanings. I will describe it using real-life examples. What is the thought that comes to your mind when you hear the word mouse, It may be the pet you keep in your home or it may be the key board that you operate your computer with. Of which one do you know what is under discussion? You scan through the words around and the general picture. When a person tells you that the mouse ran across the floor you think of the animal. When they say the mouse, they are referring to the computer device. It is this automatic process that your brain just does in a few milliseconds that we are trying to teach computers in WSD. But to humans, what appears to be so easy is in reality so complicated to machines. Take more examples to realize how deep this problem is. The book may refer to a hardcopy that one reads or serves as a reservation. Spring may be a season, a winding metallic thing, or the process of leaping. In collection of the sentence He could not bear the pain, bear is to endure, in I saw a bear in the forest, it means an animal. What is even more interesting is such words as run that has dozens of different meanings- you can run a race, run a business, run out of time or have a run in your stocking. The situation is even more complicated when we take into account the fact that the context is not always obvious. In some cases all one would do is to look at the immediate immediate surroundings words, whereas, in other cases one would have to comprehend the whole paragraph or even the general subject under discussion. Also, certain senses of words are quite similar to each other, and it is more difficult to differentiate them even in the case of human beings. At present, not a lot of digital materials or annotated data are produced on Maithili, thus hindering the further advancement. But using the concentration on the accumulation of the materials of the Maithili language, the development of small collections and the utilization of the modern language technology devices, we will be able to develop effective WSD systems in Maithili. Maithili can achieve the same level of computer-assisted language support as other major Indian languages with continued research and collaboration between the experts of the language and technologists. This demonstrates that WSD in Maithili can and would be a possibility with some effort and concern.

Word Sense Disambiguation (WSD) of the Maithili language is a chance that would provide numerous new opportunities to reinforce the digital image of the language. The process can assist in developing smart language software like translators, chatbots, search engines to the Maithili users. Given the fact that WSD determines the correct meaning of a word depending on the context, it can help to understand and communicate more effectively in an online setting. In a bid to conduct WSD of Maithili, the initial stage involves developing a set of Maithili words applied in various senses. This is achievable by gathering

materials in books, folklore, songs, and internet material. This kind of collection will assist in identifying words with multiple meaning as is the case with the word bank in English that can be used to describe a financial location or a river bank. By developing this collection, one will establish a database of Maithili words. Secondly, one should build a small Maithili WordNet or sense inventory that contains all possible meanings of words. This may be done by referring to other close languages such as Hindi and Bengali as they are similar to Maithili in grammatical and vocabularies. The independent Maithili WordNet can be created gradually and used in research and application. Maithili can use modern machine learning techniques to play a significant role in its WSD. Models can be trained using algorithms such as supervised, semi-supervised and unsupervised learning based on the available text data. They can be used with even small dataset because they can use data augmentation and transfer learning with resource-rich languages. This in turn makes the system automatically learn how the meaning of a word may vary in various circumstances. Maithili is a very enriching language in grammar and morphology and hence it makes a good study of linguistics. Through the use of WSD approaches, the researchers are able to resolve the issue of word ambiguity as well as developing the language digitally. The linguistic analysis, development of the resource and machine learning will make WSD in Maithili more precise and effective over time. Therefore, the given work may become an important step toward making Maithili a part of the realm of computational linguistics.

Table I
AMBIGUOUS WORDS

S.No	Words	Sense
1	गाछ (ganch)	i. गाछ (Tree) ii. कपड़ा गाछब (To dry)
2	बाँध (baandh)	i. जलक बाँध (Dam) ii. बाँधब (To tie)
3	हात (haat)	i. शारीरिक अंग (Hand) ii. हाथ लगाना (To help)
4	काँच (kanch)	i. काँच (Glass) ii. काँच (Raw)
5	दाम (Daam)	i. मूल्य (Price) ii. रस्सा (Rope)

II. RELATED WORK

Aarti Purohit and Kuldeep Kumar Yogi (2025) Proposed the research addresses Word Sense Disambiguation (WSD) in the Hindi agriculture field by developing a novel sense tagged corpus. It discovered that the best machine learning models in terms of accuracy were the Random Forest and SVM machine learning models of up to 87 percent. The most crucial features that the study found to be critical to success were POS tagging, word context, and n-gram probabilities[1].Michele Bevilacqua et.al. (2021) Uses the Word Sense Disambiguation (WSD) to find the correct meaning of a word in a context with the help of such tools as WordNet. Relation and supervised models based on glosses, relations and external knowledge have attained state of the art performance with results exceeding 80 percent of inter-annotator agreement. Even with the

current developments in the field of transfer learning, the issues of WSD such as insufficient benchmark and multilingual coverage remain [2]. Mosima Anna Masethe (2022) et.al. Proposed the paper addresses the issue of decoding word meanings in the highly complicated Sesotho sa Leboa language, which has a great variety of meanings. It relies on a context and meaning-based system of the words with the application of Word2Vec and SBERT that attempts to find the most suitable match. SBERT performed best, with a similarity of 87% indicating that it has the ability to identify words by their correct meaning[3]. Dr. D.K. Tayal et.al. represents a survey of the historical techniques of WSD. Recent techniques of supervised, semi-supervised, and knowledge-based techniques perform better on benchmark datasets[4]. Examples of techniques such as BERT and Z-reweighting have been demonstrated to be more effective[4]. Nasrin Taghizadeh and Hesham Faili (2016) suggested an Expectation-Maximization algorithm which can build large-scale and high-quality wordnets in low-resource languages with only a bilingual dictionary and a monolingual corpus, through cross-lingual word sense disambiguation. In a test on the Persian language, it is observed that the induced wordnet is 90 percent precise and 35 percent recalling[5]. Vinto Gujar et. al. (2023) where the reviews of Hindi Word Sense Disambiguation (WSD) techniques are presented. Knowledge-based methods based on resources such as Hindi WordNet are 40-70% accurate, supervised methods based on classifiers 79-89% and unsupervised methods based on context and embeddings 43-80%. Such gaps are a limited number of datasets, noun oriented, and idioms. Hybrid models that would integrate approaches are likely to enhance performance[6]. Hlaudi Daniel Masethe et.al. (2023) introduces the paper and identifies the role of WSD in NLP and its issue in low-resource languages. It examines supervised, unsupervised and knowledge based methods. A meta-analysis of 32 studies indicates a moderate level of effectiveness with high variability ($I^2 = 82.29$) and adds that accuracy is enhanced by bigger sample sizes. The solution to the problem of data scarcity and linguistic complexity is important in enhancing WSD[7]. Manish Sinha et.al. (2004) presents an early automatic WSD system for Hindi using the IIT Bombay Hindi WordNet. Their algorithm disambiguates nouns by matching sentence context with semantic information from the WordNet, reporting 40-71% accuracy depending on the text. The paper notes present limitations—such as coverage mainly for nouns—and points to future extensions for verbs and richer morphological analysis [8]. Ashweta Vikram and Sanjay K. Dwivedi (2019) Presents This paper reviews and compares WSD for resolving word ambiguity in digital texts and improving understanding and translation. Approaches are classified as knowledge-based or machine learning-based. Knowledge-based methods like Lesk, Selectional Restriction, and conceptual density show 44–60% accuracy but are limited by dictionary coverage[9].

Zankhana B et.al. (2024) Presents the review of WSD in Indian languages reports accuracies of 75–90% for Hindi, 52.35% F-measure for Malayalam, 85.71% F-measure for Assamese, and 63.71–80.82% for Bengali. Challenges include limited annotated data, incomplete resources, varied context, and focus on Hindi, with knowledge-based and unsupervised methods commonly used to address these issues[10]. Anup Kumar Barman et.al. (2024) Proposed this study applies WSD to improve Assamese-Hindi statistical machine translation, integrating Hindi WordNet to resolve ambiguous words before translation. Using supervised learning (Naive Bayes), the system refines translation accuracy and demonstrates that embedding WSD into MT output yields better results than a baseline system. The work is significant for low-resource NLP, especially for computationally under-resourced languages like Assamese [11]. Rasika Ransing and Archana Gulati (2023) Proposed the paper focuses on Marathi Word Sense Disambiguation (WSD), pointing out that sense-annotated corpora are very few. To address the issue, the authors have created a sense-annotated Marathi corpus and tested various supervised classifiers like Naive Bayes, SVM, Decision Tree, Random Forest, and Logistic Regression. Naive Bayes was the most effective method with precision 0.39, recall 0.53, accuracy 0.53, and F1-score 0.43. The authors have justified the poor performance due to the small dataset, which they believe can be enhanced with more data[12]. Sunita Rawat and Manoj Chandak (2015) Proposed the paper reviews NLP, machine translation, and WSD, comparing knowledge-based and machine learning-based approaches. For supervised WSD, SVM achieves the highest accuracy, while decision trees are the fastest. Among unsupervised methods, SOM has the highest accuracy, though hierarchical clustering catches up as clusters increase. K-means suits large datasets, and hierarchical clustering is better for small datasets[13]. Roshan R. Karwa and M.B. Chandak (2014) Presents paper reviews WSD methods—knowledge-based, supervised, and unsupervised—highlighting their pros and cons. It suggests a hybrid, partially annotated approach to improve accuracy and overcome existing challenges in WSD[14]. Tawseef Ahmad Mir et.al. (2023) Presents the paper introduces KLS-WSD-Corpus, the first sense-annotated dataset for Kashmiri, with 9,673 instances of 60 ambiguous words. Supervised methods like k-NN, Decision Tree, and SVM were tested, with SVM using BoW and PoS features achieving the highest accuracy of 75.74%. This dataset provides a key resource and baseline for Kashmiri WSD research [15]. Shantanu Kumar and Narayan Choudhary (2025) reflects the current state of technologies for Maithili, an Indo-Aryan language spoken mainly in Bihar, Jharkhand, and Nepal. While a few progressive steps have been taken, the availability of resources and research in the Maithili language processing domain is still scant. The document deals with various areas of language technology encompassing corpus creation (text and speech

corpora), speech recognition and synthesis, morphological analyzers, transliteration tools, language standardization issues, and machine translation attempts. It reveals that the absence of a standardized orthography and the scarcity of datasets are the biggest hurdles, and thus a lot of work is suggested for spelling tools, natural language parsers, sentiment analysis, named entity recognition (NER), wordnet creation, and optical character recognition (OCR) to be done in the future. The paper acts as a guide to the research landscape and the missing pieces to Maithili language computational development[16]. Raza Rahi et.al. (2020) Provides the article is reporting the Morphological Analyzer for Maithili with a high degree of accuracy, the test was done on 500 words per part of speech category from the esamaad corpus. The accuracy was between 92% for nouns and 98% for pronouns and adverbs. The system formally recognized 15,246 out of 16,572 nouns, 14,312 verb forms, and 593 out of 634 adjectives. As there is no comparative Maithili analyzer, the results indicate a strong level of performance in general, for different categories[17]. Boshra F et.al. (2014) Presents the paper applies a supervised decision tree (C4.5) approach to WSD using the Senseval-3 dataset. It classifies ambiguous nouns and verbs based on context, achieving an overall accuracy of 45.14%. While some words were classified accurately, results varied, showing that decision trees are useful but need further refinement for better performance [18].

III. WORD SENSE DISAMBIGUATION IN MAITHILI APPROACH

A. Rule-Based Methods

The initial phase is the development of rules in accordance with grammar and sentence structure of Maithili. Such rules consider the words that come before and after a word that is not clear, the type of the word (such as noun or verb) and structure of the sentence. An illustration is that one rule might be that, when a particular word is close to the words in question, they are more likely to have a particular meaning than the other. Rule-based systems would assist us initially when there are not many examples to train a computer.

B. Machine Learning Methods

In case we have some labeled examples, machine learning can help to enhance WSD. The training of machines is done with numerous examples of words with different senses. The model acquires patterns within the context in order to determine the right meaning. The most widespread ones are decision trees, support vector machines, and neural networks. In case the number of labeled examples is very small, one can employ semi-supervised learning with a little bit of labeled data and much unlabeled data. Transfer learning can be applied as well; it applies the models that have been trained on other similar languages such as Hindi and can be applied in Maithili. Such techniques assist the system in learning using more

data and are not constrained by a small number of labelled examples.

C. Statistical Approach

It is a method of determining the correct usage of a word by means of math and probability, which can determine how frequently words occur in a sentence. It examines trends such as the proximity between words and the nature of words that precede or follow the ambiguous word. We may resort to such models as decision trees or support vector machines. These models are based on cases where the meanings are already known. However, by simply having a limited number of these examples, statistical techniques will be able to do a great job by paying attention to sensitive aspects of the language.

D. Deep Learning Approach

Deep learning involves computer programs known as neural networks that have the capability of learning a lot about lots of text. These networks derive the meaning of words depending on the sentence. New models such as BERT and LSTM perform excellently since they are able to learn intricate word-word connections. In the case of Maithili, we can train such models using the texts we are collecting or use the models that have been trained using the similar languages such as Hindi and modify them accordingly to Maithili. Deep learning approaches improve with increased data, and are able to deal with problematic cases where meanings rely on subtle hints.

IV. METHODOLOGY

A. Data Preprocessing

Word Sense Disambiguation (WSD) is a very important part of Natural Language Processing (NLP) which deals with human language. The main purpose beyond the study is to determine the exact meaning of a word which varies according to the context in which it is used. Maithili is an Indo-Aryan language, and a considerable number of Maithili words are polysemous. Generally, people are capable of selecting the right meaning of the word from the sentence, but it is a very challenging issue for computers. Correct WSD can be a great source of the different usages of the languages such as machine translation, information retrieval, and sentiment analysis because if the meanings of the words are wrong, the system may deliver the wrong outputs.

Performing WSD for Maithili is a tougher job due to the fact that it is a language with limited resources. There are scarce annotated datasets, WordNet resources, and NLP tools for Maithili. This paper is intended to review the research done on Word Sense Disambiguation for Maithili and examine the methods employed so that it can help the next research and development in this field.

As an illustration, the Maithili term "मौसक" (mausak) may indicate either "weather" or "season" according to the sentence. The task is to input a Maithili sentence

and output the meaning that fits in the context of the ambiguous word.

B. Data Collection and Preparation

Due to the low number of digital resources for Maithili, data has to be fetched from various sources such as Wikipedia, news, blogs, and books. The text needs to be cleaned and made ready by error removal and proper formatting. Next, a group of frequently ambiguous words is selected, and native Maithili speakers label the sentences with the correct meanings. One can create a list of words with their meanings (sense inventory), for instance, by utilizing the IndoWordNet.

C. Preprocessing

The content is divided into words (tokenization), and each word is given its grammatical category (like noun, verb). There are tools like the Indic NLP Library or Stanza that can do these operations. Additional work like stop word removal and finding the root form of words (lemmatization) might also be accomplished.

D. Feature Extraction

Features are derived from the input data, depending on which method is used: Conventional models consider nearest words, part-of-speech tags, and word combinations in the context of a sentence. Also, word representations trained on Maithili materials may be employed.

E. Model Training

In the case of supervised learning, if there is already a dataset with labels, one can train models like logistic regression, SVM, LSTM, or a BERT variant that has been fine-tuned to predict meanings. When there is no labeled data, one can resort to unsupervised techniques such as clustering of similar usages of words.

Methods that rely on knowledge can employ IndoWordNet and an algorithm like Lesk to determine meanings by using definitions from dictionaries.

F. Evaluation

Performance of the system is gauged on fresh Maithili sentences that are not part of the training set. The accuracy, precision, recall, and F1-score are some of the metrics used to quantify the effectiveness of the WSD system. Also, cross-validation can be performed if there is sufficient data.

V. CONCLUSION

In the review paper, we have addressed the valuable task of Word Sense Disambiguation (WSD) of the Maithili language, a low-resource language that has disadvantages in terms of tools and data. Nevertheless, we have observed that it can be possible to create successful WSD systems with the help of rule-based approaches, machine learning, and deep learning approaches. Gathering quality Maithili

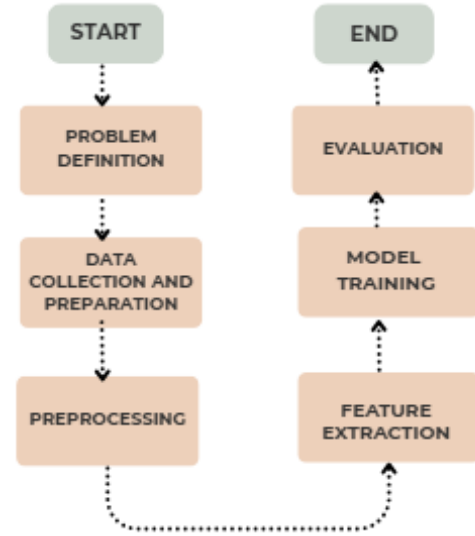


Figure 1. Algorithm for Word Sense Disambiguation (WSD)

literature and marking down on ambiguous words is important towards educating correct models. Recent models of the language such as BERT and such techniques as transfer learning between related languages can be used to enhance performance of WSD. Correct WSD of Maithili will help in improving machine translations, information search, and other language applications, which will help in the development of digital tool to Maithili. The objective of this review is to inform as to the future research directions and invite the development of more sophisticated language technologies to work with Maithili.

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